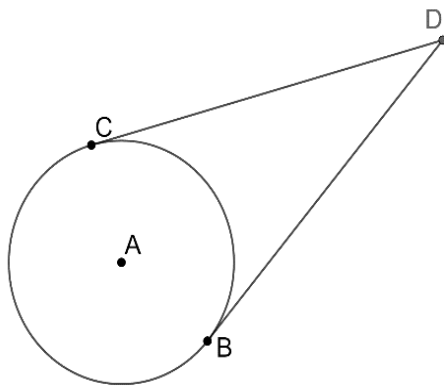


Geometry
Unit 12
Lesson 4 Practice

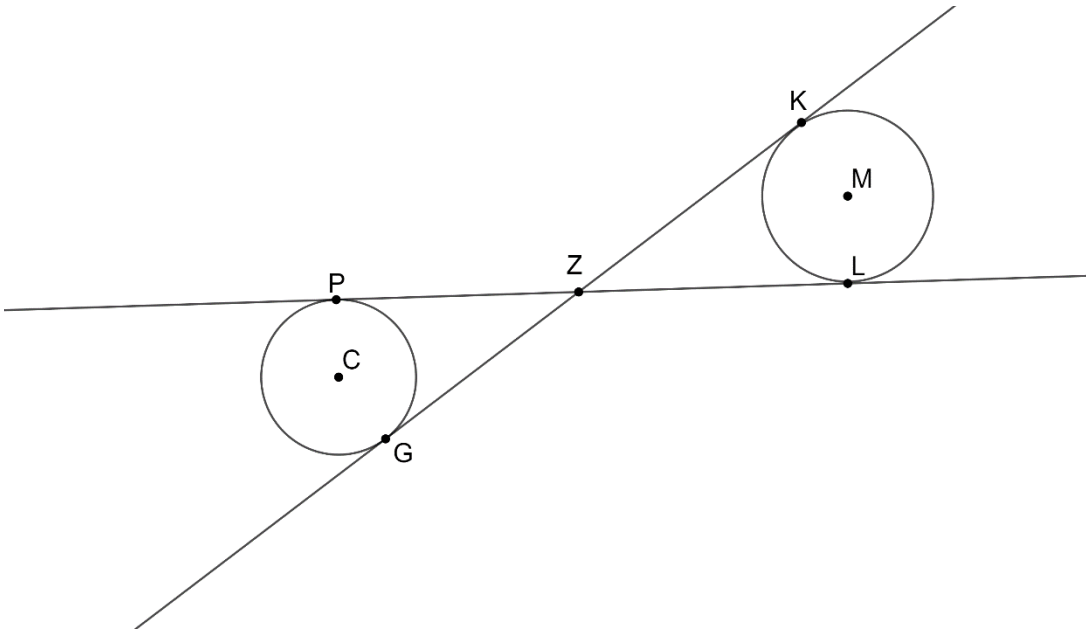
Name _____
Date _____
Period _____

For questions 1-3, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.
 $CD = 8x - 20$ and $DB = 13x - 45$.



1. What is the value of x ?
2. What is the length of $\overline{CD}$?
3. What is the length of $\overline{DB}$?

For questions 4-10, Circles C and M are shown. Lines $\overline{PL}$ and $\overline{GK}$ intersect at point Z . Line $\overline{PL}$ is tangent to circle C at point P and circle M at point L . Line $\overline{GK}$ is tangent to circle C at point G and circle M at point K . $PZ = 10y - 3$, $ZL = 4x + 10$, $GZ = 7y + 21$, $ZK = 6x - 16$.



4. Write an equation you can use to solve for x .
5. Solve the equation you wrote in question 4 for x .
6. Write an equation you can use to solve for y .
7. Solve the equation you wrote in question 6 for y .
8. What is the length of $\overline{GK}$?
9. What is the length of $\overline{GZ}$?
10. What is the length of $\overline{ZL}$?